

AT-FSV / ATAI

CMFSB ISOLATION MODULE — DOCUMENT 00

MASTER ENGINEERING BASELINE | ATFSV-MOD-001 | Captured Mechanical Floating Safety Body

Revision 5.8 | Contract Diligence Baseline | Option + Deliverables + REQ-M21 Freeze | 2026-10-03

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Field	Value
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Status	Contract Diligence Baseline — T1 review / Option negotiation; not certified; TRL 2–3

Release statement. This revision is a **Contract Diligence Baseline** for controlled T1 review, supplier evaluation, and Path A Option negotiation. **DOC-00 is not a performance certificate, type approval, measured comfort claim, or production commitment.** Certification, safety credit, and ride-comfort percentages require configuration-managed evidence outside this document.

1. Executive Summary, Value and Document Control

Non-claims. This baseline does not claim production readiness, regulatory approval, certified ASIL, crashworthiness certification, measured ride-comfort improvement, guaranteed 7,500 kg vehicle compliance, final supplier mass/cost/lead time, patent novelty, or freedom-to-operate (FTO). Those claims require controlled evidence, counsel and programme approvals identified in this document.

1.1 Value statement and planning KPIs

Market problem. Occupant compartments in specialty, defence-utility, ambulance and extreme limited-series vehicles receive road vibration that degrades comfort, task performance and, where relevant, clinical stability. Soft isolation alone can reduce handling stiffness in rapid manoeuvres. CMFSB addresses both by keeping the upper body mechanically captured at all times, providing a COMFORT isolation envelope and a SPORT mechanical lock for handling, with passive support on power loss.

Measurable advantage is not claimed numerically until L1/L2 evidence exists. Planning KPIs below are method definitions only.

KPI	Metric definition	Method	Reference	Status
Ride isolation	RMS / peak upper-body acceleration in 0.1–0.5 / 0.5–2 / 4–12.5 Hz	SIM-001 then bench	ISO 2631-1 bands; ISO 8608 roads	OPEN until L1
Handling stiffness	Roll angle / lateral response SPORT vs COMFORT under same step	SIM-001 + mule	Defined lateral proxy	OPEN until L1
Mode time	SPORT↔COMFORT complete transition	Bench / HiL	≤3 s TARGET class	TARGET
Power-loss	Settle on passive support within envelope; valves fail-closed	Power-cut test	S4 invariant	BASELINE rule
Module mass	Core tower ≤55 kg TARGET; full corner STUDY	Weighing	§4 module table	TARGET / STUDY

1.2 Disclosure tiers T0 / T1 / T2 / T3 (aligned with document tree)

Definitions below match the controlled document tree and supersede any shorter paraphrase.

Tier	Content	Access condition
T0	Public-facing non-enabling concept / teaser material	Public or first contact
T1	This confidential engineering baseline: architecture, requirements, risk, verification planning	NDA

Tier	Content	Access condition
T2	Models, detailed ICD, test data, commercial conditions and configuration-controlled design artefacts for detailed design and verification	NDA + need-to-know
T3	Patent claim charts, prior-art mapping, negotiation strategy, counsel-controlled disclosure	NDA + counsel + restricted list

Interfaces to ICD-001, SIM-001, STR-001 and REG-001 are maintained through the document tree.

1.3 Reference application A and dual running gear

Application A (planning) uses an upper-body range of 1,500–2,500 kg before occupants where an ambulance-class cabin is considered; EN 1789 and EN 1865 are planning references only. The module interface is valid for both 4×4 and 6×6 lower vehicles within the stated load classes; a 4×4 pilot is generally the shorter integration path for a first contract unless a buyer funds 6×6.

1.4 Architecture summary

CMFSB is a mechanically captured isolation module. The upper body remains geometrically constrained in every operating state. Magnetic/Lorentz force is a bounded dynamic-control layer and is never the sole static load path. The reference architecture combines a compliant hydraulic macro stage (± 50 mm body-relative) with a Lorentz micro stage (± 7.5 mm stroke; coil–magnet gap 2–5 mm baseline). COMFORT releases the SPORT lock and uses the full macro envelope with micro control. SPORT seats and mechanically locks the upper body to increase roll/yaw stiffness. Power loss transitions to S4 with passive support; leveling valves fail-closed. Evidence ladder: L0 architecture → L1 multi-band SIM (§8) → L2 single-tower bench → L3 four-corner/HIL → optional L4 mule. No measured comfort percentage is claimed.

Path A remains the primary commercial route: field license or supply of ATFSB-MOD-001 without requiring the buyer to adopt Appendix A powertrain or GVW figures. Path B is a full-vehicle or limited-series partnership that may use the carbon-cell and dual 4×4/6×6 notes from prior baselines as STUDY options. Dual-mode COMFORT/SPORT is intended to answer the classical objection that soft isolation degrades emergency handling: when SPORT is selected and locks confirm, the upper body and lower structure share a high effective stiffness path for roll and yaw while capture and progressive stops remain the crash backbone.

Supplier engagement under T1 should focus on: (1) hydraulic macro stage and fail-closed leveling valves; (2) Lorentz actuator thermal and force capability against the 0.7 kN RMS and ± 2.5 kN peak screening numbers; (3) lock face geometry and status sensing; (4) progressive-stop energy curve ownership under STR-001; (5) ICD completeness for mechanical, electrical, communication and safety. Numeric comfort claims are forbidden in external materials until SIM-001 and/or L2 records exist under configuration control.

1.5 Document control

Field	R5.8 value
Document ID / Revision / Date	ATFSV-DOC-00-R5.8 / 5.8 / 2026-10-03
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Technical Reviewer	Independent engineering reviewer — to be appointed
Approver	Not approved for road use, production or certification credit — CONTROLLED
Contact for formal assignment / buyer diligence	Via Legal Owner; corporate assignment deed to be executed before entity-level sale if required

1.6 Status vocabulary (exclusive)

Label	Meaning
CONTROLLED DESIGN LIMIT	Architecture ceiling; not a measured result
BASELINE	Included in current architecture accounting
TARGET	Numerical target requiring verification
STUDY	Study allocation; not frozen design
ASSUMPTION	Modelling assumption needing confirmation
DERIVED	Calculated from stated assumptions
OPTION	Not credited in baseline safety or mass closure
OPEN / TBD	Unresolved or not responsibly fixed at this revision

2. Engineering Baseline and System Architecture

Static load is carried by compliant mechanical/hydraulic support. Lorentz actuation is a bounded dynamic-control layer. Module parameters are primary. Vehicle-level mass and energy figures in Appendix A are reference-only and are not prerequisites for Path A module licensing.

Parameter	R5.8 value	Status
Architecture	Mechanically captured upper body + macro–micro isolation (NOT free levitation)	BASELINE
Modes	COMFORT (isolation) + SPORT (mechanical lock)	BASELINE
Topology	Hybrid hydraulic macro + Lorentz micro; PM+EM OPTION only	BASELINE / OPTION
Macro / micro stroke	± 50 mm body-relative (100 mm range) / ± 7.5 mm	TARGET
Lorentz gap	2–5 mm coil–magnet baseline; 15–30 mm only for PM+EM OPTION	TARGET
Peak / RMS force	± 2.5 kN study per tower / ≈ 0.7 kN RMS planning	STUDY
Km	≥ 63 N/ $\sqrt{W}$ ASSUMPTION — FEA / supplier	ASSUMPTION
Mean copper loss (4 towers)	≤ 0.5 kW continuous class at 0.7 kN RMS	TARGET
Supply	48 V class nominal (final ICD)	STUDY
Core tower mass	≤ 55 kg TARGET (structure + Lorentz + sensors + local electronics)	TARGET
Static support floor	≥ 6.5 kN per corner planning	STUDY
Leveling	Active; fail-closed valves on power loss	TARGET
Hold function	Optional rate-limited hydraulic orifice or friction device; not primary crash path — see §2.4	BASELINE optional

The reference topology remains hybrid hydraulic macro plus Lorentz micro. Active hydraulic-only macro is an allowed study path for leveling and large-stroke support. PM+EM reluctance devices remain OPTION and may not be credited in baseline mass or safety closure until a stability gate and force-gap evidence exist. Carriage first structural mode shall be at least three times the micro control bandwidth target so that structural resonance does not masquerade as control authority.

Leveling valves are fail-closed on loss of command or supply so that a power-loss event retains height rather than dumping the upper body through the full macro stroke. Progressive stops and the capture frame are sized for residual motion after passive settle. Human authority is preserved: driver and rescue retain physical access paths, and S8 maintenance lockout is a physical device, not a software flag alone.

2.1 Safety invariants

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture
Static support	Vehicle weight on passive or fail-safe compliant paths — never sole magnetic path
Power loss	S4 passive support; leveling valves fail-closed; software aliveness not required
Crash path	Progressive stops + structure; independent of closed-loop magnetic force
Mode transition	SPORT $\leftrightarrow$ COMFORT never enters unconstrained free-float

Invariant	Rule
SPORT lock fault	Lock failure shall not defeat capture or progressive stops
Horizontal restraint	Longitudinal/lateral/yaw loads resolved by dedicated paths and/or SPORT lock — see §2.5 and REQ-M25

2.2 Functional architecture

Layers (top→bottom): Upper occupant/safety body → Micro stage (carriage + Lorentz) → Macro stage (hydro-pneumatic + leveling + lock seats) → Lower chassis. Magnetic subsystem is never the sole vertical load path.

2.3 Mode definition

Mode	Mechanical state	Objective
COMFORT	Locks released; macro ± 50 mm; micro active; leveling active	Soft ride isolation
SPORT	Upper body seated; locks confirmed; isolation stages held	Handling stiffness (roll/yaw)

Selection paths (planning): driver command (highest explicit priority); radar/surface-based proposal; GPS/map-based proposal. Arbitration rules, confidence thresholds and lateral-g inhibit of unlock remain OPEN until vehicle HMI/dynamics freeze. Default after self-test is COMFORT unless ICD states otherwise.

2.4 Hold function (definition for REQ-M16)

Hold is an optional secondary rate-limited restraint, distinct from the SPORT positive lock and from progressive stops. Implementation class: hydraulic orifice and/or friction device at the macro stage. Engagement is rate-limited (velocity and force limits OPEN — freeze in ICD). Hold is not a primary crash energy path; progressive stops remain ultimate. Mass of hold hardware is included in corner hydraulic/stop share. Failure modes: unwanted engagement (plausibility + rate limit) and failure to engage (no safety credit required). If hold is not fitted, REQ-M16 is marked N/A for that configuration.

2.5 Horizontal restraint (longitudinal / lateral / yaw)

Macro travel ± 50 mm and progressive stops are specified primarily in the vertical axis. Braking, cornering and crash introduce horizontal loads. For an occupied upper body $\approx 2,650$ kg: 1 g brake or lateral ≈ 26 kN total; 20 g horizontal crash class ≈ 520 kN total. These loads shall not rely on Lorentz force. Paths: (1) capture frame and linkage shear capacity; (2) dedicated horizontal stops or links with defined free play; (3) SPORT lock faces providing additional shear/roll restraint when engaged. REQ-M25 owns numeric free play, stiffness and load share. OPEN until STR-001 and ICD freeze geometry.

3. Travel, Stiffness, Leveling, Clearance Stack and Actuator Screening

3.1 Frequency, deflection and leveling

Corner mass share (occupied $\approx 2,650$ kg / 4) ≈ 662.5 kg. At $k \approx 40$ kN/m, $f_n \approx 1.23$ Hz. Payload change $\Delta F \approx 2.4$ kN yields ≈ 60 mm static deflection at 40 kN/m ($\approx 60\%$ of 100 mm total macro range), leaving $\approx \pm 20$ mm dynamic stroke without leveling — insufficient for isolation targets. Active leveling restores the operating point. Air-spring progressive rate helps but does not remove the need for controlled leveling.

3.2 Clearance and stroke stack (nominal → stop)

Terms shall not be used interchangeably. The stack is: (1) nominal operating height set by leveling; (2) macro travel envelope ± 50 mm about that height; (3) residual free play / pre-trigger gap before progressive-stop contact (planning example 30–50 mm depending on mode and S5 preparation); (4) progressive-stop stroke and force–displacement curve (STR-001).

Segment	Planning description	Example sensitivity
Nominal height	Leveling set-point under stated payload	Shifts with payload if leveling fails
Macro envelope	± 50 mm body-relative about nominal	100 mm total range

Segment	Planning description	Example sensitivity
Pre-trigger / free play to stop	Gap after macro limit before stop energy rise	If $s = 30$ mm and $a \approx 20$ g class approach, $v = \sqrt{(2as)} \approx 3.43$ m/s; $E \approx \frac{1}{2}mv^2 \approx 15.8$ kJ for 2,650 kg
Stop stroke	Energy absorption displacement (STR-001)	Curve OPEN — not a compliance claim

S5 (impact preparation) may reduce effective free play by pre-positioning; energy and velocity numbers must be recomputed for the frozen S5 geometry. Until STR-001, all energy figures are STUDY.

3.3 Lorentz and optional PM+EM

$K_m \geq 63$ N/ $\sqrt{W}$ is ASSUMPTION. At 0.7 kN RMS, copper loss per tower ≈ 123 W $\rightarrow \approx 0.5$ kW for four towers (aligned with mean-power TARGET). Peak 2.5 kN $\rightarrow \approx 1.57$ kW/tower pulse; ≈ 6.3 kW / ≈ 131 A at 48 V for four towers — ICD must size drivers and wiring. Continuous thermal design is dominated by copper loss; mechanical power in the 4–12.5 Hz ride band is typically tens of watts. PM+EM reluctance is OPTION only; 15–30 mm working gap applies only to that option.

3.4 Power budget boundary

Mean continuous copper-loss class ≤ 0.5 kW (4 towers) at planning RMS. Peak pulse class ≈ 6.3 kW copper. Module 48 V supply is independent of any vehicle HV system. In SPORT, Lorentz continuous demand is near zero (stages held).

4. Mass Architecture — Module Table, Vehicle WBS and Loads

4.1 Module mass breakdown (licensable object — line visible)

The ≤ 55 kg TARGET applies to the core tower only. Full corner and shared hydraulics are separate STUDY lines so the licensed module is traceable in the WBS.

Element	P50 kg	P90 kg	Scope
Local structure (per tower)	18	25	Core
Lorentz + local carriage	22	30	Core
Sensors + local electronics	8	12	Core
Core tower subtotal	48	67	≤ 55 kg TARGET at P50; P90 OPEN
Cylinder + accumulator share	15	22	Corner
Valves / hoses / seals	5	8	Corner
Progressive stops	8	12	Corner
SPORT lock share	6	10	Corner
Harness / dampers / bellows	4	6	Corner
Pump + reservoir share (1/4)	8	12	Shared
Full corner assembly	≈ 94	≈ 137	STUDY — not core TARGET
Four corners total	≈ 376	≈ 548	Maps to vehicle WBS isolation line

4.2 Cabin / upper-body mass states

State	Mass	Static / tower
Empty upper (ref.)	2,150–2,350 kg	≈ 5.3 – 5.8 kN
Occupied 4 $\times$ 100 kg	$\approx 2,550$ – $2,750$ kg	≈ 6.3 – 6.8 kN
Maximum study	$\approx 2,950$ – $3,150$ kg	≈ 7.2 – 7.7 kN
App. A cabin class	1,500–2,500 kg	≈ 3.7 – 6.1 kN

Statistical treatment of P50/P90 in this baseline is a simple line-sum study, not a Monte-Carlo mass system. Line-sum P90 is intentionally conservative. Where a single component has a supplier-quoted distribution, that quote supersedes the study line after formal change control. Occupied mass adds declared occupant and cargo policy; the 4 $\times$ 100 kg case is the primary handling and isolation design case unless Application A defines a different seated mass.

4.3 Bottom-up vehicle mass WBS (STUDY — not supplier data)

Upper empty body $\approx 2,250$ kg mid is the sum of structure, closures, glass, interior, seats, belts, HVAC and cabin electrics — not a single “body/cabin 1,200 kg” line. Powertrain lines below are a consistent reference hybrid study set; they are not a second contradictory configuration. All figures STUDY.

WBS line	P50	P90	Note
Upper structure + closures + glass	1,100	1,200	Part of $\approx 2,250$ empty upper
Interior + seats + belts + HVAC + cabin LV	1,050	1,150	Completes empty upper $\approx 2,150$ – $2,350$
Lower chassis + crossmembers	650	750	Chassis
ICE + accessories (study class)	350	430	Aligned prior hybrid study
Generator + power electronics	220	300	Powertrain
Traction motors + inverters + reductions	450	550	E-drive
Battery / REESS ≈ 130 kWh class	700	850	Energy storage
HV/LV distribution + cooling	200	260	Electrical
Axles / steering / foundation suspension	450	520	Supplier data required
Wheels + tyres + brakes	400	480	High-speed rating OPEN
Isolation 4 $\times$ full corner (from §4.1)	376	548	Module-visible line
Fluids + CTIS + exhaust + fillers	120	160	Vehicle
Programme reserve	150	200	Held outside owners
Curb no fuel	$\approx 6,216$	$\approx 7,398$	STUDY line-sum
Fuel ≈ 390 L @ 0.75	293	293	Explicit
Curb + fuel	$\approx 6,509$	$\approx 7,691$	P90 + occupants stresses 7.5 t GVW

P90 curb+fuel plus occupants can exceed 7,500 kg operational GVW. Options: mass reduction, raise GVW with tyre/brake/homologation review, or reduce energy content. Label remains STUDY — not supplier-weighted data.

Traceability rule: any external quotation of module mass shall cite either the core-tower TARGET (≤ 55 kg) or the full-corner STUDY total, never a blended number that hides hydraulic and lock content. Vehicle WBS isolation line must equal four full corners within rounding. If a supplier quotes a different split, the WBS is updated by formal change — not by silent overwrite of the core TARGET.

Empty upper-body mass $\approx 2,150$ – $2,350$ kg is an accounting identity: structural shell and closures plus glass plus interior, seats, restraints, HVAC and cabin electrical. Prior single-line “body/cabin 1,200 kg” presentations under-counted interior and safety content by approximately one tonne and are superseded by the two-line upper split in §4.3. Battery 700/850 kg and ICE 350/430 kg restore consistency with the ≈ 130 kWh hybrid study used in earlier controlled baselines; inflated ICE or deflated battery figures that held a fixed total without physics rationale are rejected.

4.4 Reference load cases and per-tower crash note

Case	Basis	Planning magnitude
Operational 3 g vertical	3 $\times$ static class / tower	S/M/L dependent; L-class order 22–24 kN
Crash vertical 10–20 g	Design mass / 4 before asymmetry	Order 65–130 kN/tower L-class on 2,650 kg
Asymmetry factor 1.5–2	3-tower or uneven stop share	≈ 225 – 300 kN class possible — STR-001 + R-08
Horizontal 1 g / 20 g	2,650 kg upper	≈ 26 kN total / ≈ 520 kN total — REQ-M25
SPORT lock lateral	Manoeuvre + uneven lock	OPEN — ATFSV-LCK-001

A screening 150 kN/tower vertical before asymmetry is a STUDY floor only. With 1.5–2 redistribution, 225–300 kN class must be visible in STR-001 and in risk R-08 given core tower ≤ 55 kg TARGET.

5. Controlled Requirements and Safety State Machine

5.1 Unified module requirements

ID	Requirement	Owner	Verify	Gate	Status
REQ-M01	Static load capacity per size class S/M/L	Structures	Static bench	G2	TARGET
REQ-M02	Macro stroke ± 50 mm without clash	Mechanical	CAD + bench	G1	TARGET
REQ-M03	Lorentz ± 7.5 mm; gap 2–5 mm baseline	Actuator	Calibration	G2	TARGET
REQ-M04	Bandwidth ≥ 20 Hz; carriage mode $\geq 3 \times$ BW	Control	Bench dynamics	G2	TARGET
REQ-M05	Peak ± 2.5 kN; RMS ≈ 0.7 kN planning; Km stated	Actuator	Force + thermal	G2	STUDY
REQ-M06	Core tower mass ≤ 55 kg TARGET	Module	Weighing	G2	TARGET
REQ-M07	Power-loss $\rightarrow$ S4 passive; valves fail-closed	Safety	Power-cut	G2	BASELINE
REQ-M08	Multi-band SIM-001 acceptance per §8	Sim	SIM report	G1	TARGET
REQ-M09a	Progressive-stop screening curve (e.g. 600 kN class study) until STR-001	Structures	FEA / test	G3	TARGET
REQ-M09b	Stops absorb crash energy without magnet loop	Structures	Impact / FEA	G3	TARGET
REQ-M10	Life $\geq 10,000$ h equivalent road duty	Reliability	Duty test	G3	TARGET
REQ-M11	Operating temperature range frozen	Thermal	Chamber	G2	OPEN
REQ-M12	EMI / cabin field plan	EMC	EMC plan	G3	OPEN
REQ-M13	ICD mech/elec/comm/safety/lock released	Systems	ICD review	G1	TARGET
REQ-M14	No single failure causes loss of capture	Safety	FMEA+FTA	G2	TARGET
REQ-M15	Manual recovery / rescue access	HF/Safety	Access trial	G3	TARGET
REQ-M16	If hold fitted: rate-limited orifice/friction; not primary crash path (§2.4)	Safety	Design+FMEA	G2	BASELINE opt.
REQ-M17	SPORT lock positive confirmation; ≤ 3 s class	Lock	Bench	G2	TARGET
REQ-M18	COMFORT full macro without lock interference	Mechanical	Stroke test	G2	TARGET
REQ-M19	Mode arbitration driver/radar/GPS with override	Controls	HIL	G3	STUDY
REQ-M20	Transition never unconstrained free-float	Safety	Transition test	G2	BASELINE
REQ-M21	COMFORT roll-gradient / lateral-g policy numeric and frozen	Dynamics	SIM+HIL	G2	OPEN
REQ-M22	Leveling preserves dynamic stroke under Δ payload	Hydraulics	Payload test	G2	TARGET
REQ-M23	Incomplete lock set inhibits SPORT declaration	Lock/Controls	HIL	G2	TARGET
REQ-M24	Fluid systems observe ISO 4413 safety rules path	Hydraulics	ICD+review	G1	TARGET
REQ-M25	Horizontal X/Y/yaw free play, stiffness, stop loads and SPORT lock shear share frozen	Structures	STR-001+ICD	G2	OPEN

5.2 S0–S8 state machine (including S2S)

State	Name	Rule
S0	Safe / Captured	No powered release
S1	Precharge / Self-test	Capture OK; 48 V path checked
S2	COMFORT isolation	Full envelope; capture mandatory
S2S	SPORT locked	All required locks confirmed; stages held
S3	Degraded	Reduced authority; force bounded
S4	Power-loss capture	Passive support; valves fail-closed
S5	Impact preparation	Prepare stops; inhibit high-rate unlock
S6	Impact / arrest	Stops; no magnet energy dependence
S7	Post-event safe	No automatic re-enable
S8	Maintenance lockout	Physical lockout

Key transitions: S2 $\leftrightarrow$ S2S on mode request with physical preconditions; S2/S2S/S3 $\rightarrow$ S4 on power loss (hardware-dominant); S4 $\rightarrow$ S1 only (forbidden direct S4 $\rightarrow$ S2). Forbidden: free-float; S6 depending on closed-loop magnet force; SPORT with incomplete locks.

6. Summary FMEA and Preliminary HARA

Failure mode	Effect	Mitigation	S	O	D
Carriage jam	Loss of micro authority	Sensors; stops; passive	9	3	4
Link fracture	Local free motion risk	Capture frame; stops	9	2	5
Leveling stuck	Stroke consumed	Level sensor; S3	8	3	3
Hose burst	Loss of support/leveling	Pressure; stops	9	3	3
Coil overheat	Derate / shutdown	Temp model	7	3	3
Magnet Km loss	Force constant drop	Temp limits; derate	7	2	4
SPORT lock fail engage	Remain COMFORT	Status sensors; alert	6	3	2
Partial lock 1–2 of 4	Asymmetric stiffness	Inhibit S2S	8	2	2
False COMFORT at high lat-g	Excess roll	Lat-g / yaw inhibit	7	2	3
Horizontal path overload	Capture shear exceed	REQ-M25; stops	9	2	4
Power loss	Active force lost	S4 invariant	—	—	—
Software / OTA fault	Bad command	Bounds; S3/S4	8	3	4

Hazard	Scenario	S/E/C cand.	Safety goal (draft)
Loss of capture	Carriage/link failure	S3/E3/C3	Maintain capture under single failure
Uncontrolled vertical motion	Force runaway	S2/E3/C2	Bound force; S3/S4
Asymmetric SPORT lock	Partial lock + lateral load	S2/E3/C2	Inhibit S2S unless complete
Horizontal overload	Brake/crash without path	S3/E2/C3	REQ-M25 paths sized
Unexpected height drop	Valve open at speed	S2/E3/C2	Fail-closed valves
Post-crash auto re-enable	Uncommanded motion	S2/E2/C2	S7 no auto re-enable

S/E/C and FMEA ratings are PRELIMINARY screening labels, not formal ISO 26262 ASIL assignment.

7. Functional Safety, Cyber, Environment and Service

E/E elements of isolation control and mode arbitration are candidates for ISO 26262 analysis. Mechanical capture, passive support and progressive stops are independent of E/E and form the primary safety backbone. Formal HARA and FTA are G2 work products. Cyber/OTA paths that change force bounds or mode policy require authenticity and integrity controls. Environment limits (REQ-M11) and cabin field plan (REQ-M12) are OPEN until frozen. Service intervals for stops, hydraulics, locks and sensors are ICD-owned. Module Path A is not a type-approved vehicle.

8. ATFSV-SIM-001 Skeleton — Evidence Ladder L1

SIM-001 is the first quantitative evidence product. This section freezes inputs, outputs and acceptance planning. No completed numerical claim is made in R5.8.

Item	Planning definition
Model class P0	Quarter-car / half-car: upper mass, macro k & c, optional micro force, SPORT = high-k constraint
Reference passive	Frozen passive hydro-pneumatic k & c stated on every plot
Road class	ISO 8608 segments; at least one rough and one smooth
Mass states	Empty / occupied / maximum as §4.2
Frequency bands	0.1–0.5 / 0.5–2 / 4–12.5 Hz RMS and peak
COMFORT policy	Roll-gradient / lateral-g numeric policy per REQ-M21 (OPEN until frozen)
Outputs	Transmissibility; band RMS; stroke histogram; roll proxy SPORT vs COMFORT; transition time history
SIM-C1	COMFORT lower RMS in ≥ 1 agreed band vs frozen passive on rough profile — TARGET after valid run
SIM-C2	SPORT higher roll stiffness / lower roll than COMFORT on same lateral step — TARGET
SIM-C3	Macro within ± 50 mm without continuous stop hammering on agreed rough profile

Item	Planning definition
SIM-C4	No unconstrained free-float interval in transition cases
Evidence ladder	L0 architecture (this DOC) → L1 SIM-001 → L2 single-tower → L3 four-corner/HIL → L4 mule optional

Execution order for SIM-001: freeze assumptions sheet (mass, k, c, reference passive, road files) → implement COMFORT vs passive → add SPORT constraint → export band tables → lateral step proxy → review against SIM-C1...C4 with configuration hash. Results marked PRELIMINARY until independent review. Software, parameter set and road file identifiers are mandatory fields on every released plot pack.

COMFORT roll-gradient policy (REQ-M21) shall state at least: maximum steady-state roll angle per g lateral at design payload; lateral-g threshold above which automatic transition to COMFORT is inhibited; and driver-override behaviour. Until those numbers are frozen, automatic mode proposals from radar or map paths shall not force a COMFORT entry during high lateral demand.

9. Gates, Risk Register and Phase Schedule

9.1 Gates G0–G5

Gate	Intent	Entry evidence (planning)
G0	Concept controlled	DOC-00 issued; legal owner recorded
G1	Interface & SIM ready	ICD draft; SIM assumptions frozen; clash check
G2	Single-tower evidence	Force/thermal/power-loss/lock bench records
G3	Four-corner / HIL	Mode transition; stops; fault injection
G4	Mule optional	ISO 2631-1 + handling structure on donor
G5	License supply readiness	Buyer-accepted package; commercial term sheet

9.2 Phase schedule Ph0–Ph3

Phase	Month (planning)	Content
Ph0	0–2	CAD envelope; lock faces; ICD draft; procurement package after ICD baseline ≈ Month 2; supplier lead 4–8 months planned
Ph1	2–5	Single-tower force, thermal, power-loss, lock cycle
Ph2	5–9	Four-corner 6-DOF; SPORT↔COMFORT ≤3 s; stops; power-loss from each mode
Ph3	9–15 optional	Donor-chassis ISO 2631-1 and handling evaluation

Schedule assumes funded programme. Without funding, dates are not commitments.

9.3 Risk register (L×I, 1–5 scale)

ID	Risk	L	I	L×I	Owner
R-01	Core tower mass >55 kg at P90	4	4	16	Module
R-02	Km ASSUMPTION not achieved	3	5	15	Actuator
R-03	Lorentz thermal / force-gap limits	3	5	15	Actuator
R-04	Macro–micro clash or carriage failure	3	4	12	Mechanical
R-05	Stroke consumed without leveling	3	4	12	Hydraulics
R-06	Regulatory path unclear	3	4	12	Homologation
R-07	Vehicle P90 exceeds 7.5 t GVW	4	4	16	Mass
R-08	Per-tower crash load 225–300 kN vs ≤55 kg core	4	5	20	Structures
R-09	Comfort unproven until L1/L2	4	4	16	Sim+PoC
R-10	EMI / field exposure cost	2	4	8	EMC
R-11	Hydraulic mass above STUDY	3	3	9	Hydraulics
R-12	Liability if safety case incomplete	2	5	10	Legal+safety
R-13	SPORT lock reliability / partial engage	3	4	12	Lock
R-14	Mode arbitration false positive/negative	3	3	9	Controls

ID	Risk	L	I	L×I	Owner
R-15	Horizontal restraint undersized (REQ-M25)	3	5	15	Structures
R-16	Clearance stack energy under-estimated	3	4	12	Structures

10. Immediate Work Package and Document Tree

Pri	Work package	Closure deliverable
G1	Freeze CAD envelope, lock faces, leveling, fail-closed valves	3-D package + clash report
G1	ICD-001 draft mech/elec/comm/safety/lock	Buyer-readable ICD set
G1	SIM-001 assumptions + first COMFORT vs passive run	Plot pack v0.1 + hash
G2	Single-tower force, thermal, power-loss, lock cycle	Bench data package
G2	Formal HARA + FTA capture and asymmetric lock	Safety work products
G2	STR-001 stop curve + REQ-M25 horizontal loads	Structures memo
G3	Four-corner rig mode transitions	6-DOF + ≤ 3 s records
G3	Mode arbitration HIL stubs	Arbitration test note
G5	Path A Option / term sheet support	Commercial pack under counsel

Document tree (controlled): DOC-00 (this baseline); MOD-001 datasheet; ICD-001; SIM-001; STR-001; LCK-001; ACT-001; REG-001; ENE-001; BRK-001; IP-001 (T3). Each child document carries its own revision and does not silently amend DOC-00 totals without a change record.

Appendix A — Reference Vehicle Context (Path B only)

Vehicle-level energy and GVW figures support Path B sizing. They are not required for Path A module license. GVW operational baseline 7,500 kg; development mule may use lower GVW. Mission energy screening at 7,500 kg remains STUDY (road load, fuel ≥ 390 L class, brake fade risk at high-speed gated development). Detailed energy tables from prior baselines remain valid as STUDY until ATFSV-ENE-001 is configuration-managed.

Item	7,500 kg operational	6,000 kg development
Purpose	Operational accounting, tyre, brake, payload	Lab / mule only
Curb practical target	$\leq 6,500$ kg programme aim	$\leq 5,000$ kg objective (not closed)
High-speed note	250 km/h gated development only — not continuous public-road claim	Same

A.1 Mission energy and brake screening (7,500 kg basis — STUDY)

Quantity	Screening value	Note
Rolling power @ 120 km/h	≈ 29.4 kW	DERIVED
Aero power @ 120 km/h (CdA 2.3 m ²)	≈ 52.2 kW	STUDY
Road load @ 120 km/h	≈ 81.6 kW	Before auxiliaries
Mission energy 1,200 km class	≈ 886 kWh mechanical/road class	STUDY — ENE-001
Fuel with 10% reserve	≈ 365 – 391 L → TARGET ≥ 390 L	DERIVED
Single stop from 250 km/h	≈ 18.1 MJ	Kinetic
Repeated 250/200/150/120 cycle	≈ 40 MJ cumulative class	Without regen — fade risk

Brake thermal model ATFSV-BRK-001 remains OPEN. 250 km/h is gated development only.

Appendix B — Path A Commercial Engagement (Planning, Non-Binding)

Recommended ladder: T0 teaser → NDA + T1 (this document) → paid Option (60–120 days, T2 access, optional limited exclusivity) → license/supply term sheet with milestones (ICD freeze, single-tower evidence, integration support, royalty/unit). Option fee amounts are bilateral and are not published as fixed public prices in this baseline. Full IP assignment for a single large upfront payment without

Option/PoC/FTO is not the baseline Path A structure.

B.1 What Path A delivers

Included under closed Path A license (examples)	Not included unless separate agreement
Rights to use defined CMFSB module architecture in a named field	Worldwide assignment of all related IP
Controlled package: ICD, datasheet, safety behaviour, dual-mode rules	Turnkey certified vehicle or type approval
Agreed engineering support hours	Unlimited R&D; for buyer entire vehicle line
Updates within a defined revision window if contracted	Guaranteed comfort % without L1/L2 evidence

B.2 Buyer actions that shorten time-to-term-sheet

Execute NDA quickly; name technical and commercial owners; state field of use; accept Option structure before demanding full assignment; co-fund or fund minimal evidence (SIM-001 or single-tower). DOC-00 alone does not replace those steps.

Appendix C — Verification, Test and Acceptance Framework

C.1 Ph0 to Ph3 acceptance

Ph0: full-stroke clash check; lock-face clearance both modes; ICD draft reviewed; procurement after ICD ≈ Month 2; supplier lead 4–8 months. Ph1: force-gap/current curve; power-loss landing; thermal RMS; lock engage/release with confirmation. Ph2: four-corner 6-DOF; SPORT↔COMFORT ≤3 s; power-loss from each mode; stop impact in envelope. Ph3 optional: ISO 2631-1 and handling on donor chassis.

C.2 L2 single-tower test outline

Force-gap/current; power-loss landing to S4; lock engage/release ≤3 s class; thermal RMS; contamination robustness; stop contact within STR-001 curve.

C.3 L3 four-tower / HIL outline

Corner coherence including asymmetry; mode transition; power-loss policy; fault injection (confirm loss, supply loss, sensor freeze, command integrity); controlled stop engagement without loss of capture.

C.4 Evidence records

Every record retains software version, parameter set, configuration hash, road/profile ID, payload, mode, passive k/c, time histories, uncertainty where applicable, review status. PRELIMINARY results stay labelled until independent review.

Appendix D — Controlled References, Glossary and Quick Reference

D.1 Planning standards

Reference	Use	Status
ISO 26262	Functional safety process candidate	Candidate
ISO 21448 (SOTIF)	Safety of intended functionality	Candidate
ISO 2631-1	Whole-body vibration evaluation	Candidate
ISO 8608	Road surface profiles for simulation	Candidate
ISO 4413	Hydraulic fluid power — general rules and safety requirements (REQ-M24)	Reference
EN 1789 / EN 1865	Ambulance planning references for Application A	Candidate
Market vehicle standards	Category-specific — buyer confirms	OPEN

D.2 Glossary (selected)

Term	Meaning
CMFSB	Captured Mechanical Floating Safety Body — not free levitation

Term	Meaning
Core tower	Structure + Lorentz + sensors + local electronics (≤ 55 kg TARGET)
Hold	Optional rate-limited orifice/friction restraint; not primary crash path
SPORT / COMFORT	Locked handling mode / soft isolation mode
G0–G5	Programme gates in §9.1

D.3 Quick reference

Architecture: captured macro-micro; no free float. Modes: COMFORT + SPORT lock. Topology: hydraulic + Lorentz; PM+EM optional. Macro/micro ± 50 / ± 7.5 mm. Lorentz gap 2–5 mm baseline. Force ± 2.5 kN study / ≈ 0.7 kN RMS planning. Static floor ≥ 6.5 kN/corner. GVW baseline 7,500 kg reference. Comfort evidence requires L1/L2.

D.4 External disclosure and FTO

T1 contains technical architecture, requirements, risk and verification planning. Patent claim charts, prior-art mapping and negotiation strategy remain T3. This document asserts neither inventive novelty nor freedom-to-operate; FTO and patentability are counsel matters outside T1.

11. Mode Transition Sequence, Lock Architecture and Horizontal Load Paths

11.1 COMFORT → SPORT sequence (TARGET ≤ 3 s under defined conditions)

Step	Action	Interlock
1	Mode request accepted by arbitration	No active inhibit (lat-g, fault)
2	Macro/leveling drives upper body to seat band	Stroke sensors in range
3	Seat contact detected at required corners	Contact flags true
4	Positive locks command engage	Actuator enable healthy
5	All required locks report engaged	Incomplete set → abort to COMFORT
6	Isolation stages held; state S2S	Driver indication ON

11.2 SPORT → COMFORT sequence

Release is inhibited above lateral-g / yaw thresholds (REQ-M21). When allowed: locks command release → all report released → macro envelope re-enabled → micro control re-enabled → state S2. Forbidden: declaring COMFORT while any required lock remains engaged.

11.3 Lock architecture planning (ATFSV-LCK-001)

Element	Function	Status
Seat faces	Upper hard-point to lower receiver at seat band	Geometry OPEN
Positive lock	Pin, claw or hydraulic clamp — prevents separation	Per-corner status required
Actuation	Hydraulic or electromechanical; may share macro power	Duty cycle STUDY
Fail behaviour	Fail to known state with driver alert; never free-float	BASELINE rule
Crash role	May share load; progressive stops remain primary energy path	BASELINE

11.4 Horizontal load path summary

Load	Order of magnitude (2,650 kg)	Path
Brake 1 g	≈ 26 kN total	Capture shear + links + SPORT lock if engaged
Lateral 1 g	≈ 26 kN total	Same; roll couples into vertical towers
Crash 20 g horizontal class	≈ 520 kN total	Structure + horizontal stops; not Lorentz
Vertical crash with 1.5–2 asymmetry	≈ 225 – 300 kN per tower class	Stops + cell; R-08

REQ-M25 freezes allowable free play in X/Y/yaw for COMFORT, the stiffness contribution of SPORT locks, and the rating of horizontal stops. Until that freeze, integration drawings shall show provisional envelopes only and shall not claim crash compliance.

12. Lorentz Screening Detail and Thermal Notes

Planning continuous operation uses ≈ 0.7 kN RMS per tower to keep four-tower copper loss near ≤ 0.5 kW when $K_m = 63 \text{ N}/\sqrt{\text{W}}$. If supplier K_m is lower, either RMS must be reduced or the mean-power TARGET revised — silent retention of both numbers is not permitted. Peak ± 2.5 kN is a study authority band for transient events, not a continuous rating. ICD shall publish winding temperature limits, derate map and allowable duty cycle after bench measurement (ACT-001).

Quantity	Value	Comment
K_m	$\geq 63 \text{ N}/\sqrt{\text{W}}$	ASSUMPTION until FEA/supplier
RMS planning	$\approx 0.7 \text{ kN} / \text{tower}$	Aligned with $\leq 0.5 \text{ kW}$ mean (4 towers)
Peak study	$\pm 2.5 \text{ kN} / \text{tower}$	Pulse thermal in ICD
Peak copper (4×2.5 kN)	$\approx 6.3 \text{ kW}$ class; $\approx 131 \text{ A}$ @ 48 V	Driver/wiring sizing
Gap baseline	2–5 mm coil–magnet	Not 15–30 mm (OPTION only)
SPORT copper demand	Near zero continuous	Stages held

13. Integration Notes for 4×4 and 6×6 Host Vehicles

The module ICD defines four upper hard-points and load classes S/M/L. A 4×4 host places conventional axle loads under the lower structure; a 6×6 host may lengthen the wheelbase and add a mid-axle without changing the four-tower upper pattern if static load per tower remains inside the selected class. Steering of additional axles, tyre rating and brake thermal capacity are host-vehicle responsibilities. For the first paid pilot, 4×4 is the default recommendation unless the Option agreement funds 6×6-specific integration.

Topic	4×4	6×6
Module interface	Four towers	Same four towers if load class held
Integration risk	Lower	Higher (axle, steer, packaging)
First contract bias	Preferred pilot	Second SKU or funded path
Differentiation	Road / limited series	Extreme / specialty / utility

14. Non-Claims Register and External Communication Rules

External T1 readers and sales conversations shall treat the following as mandatory non-claims unless a later configuration-managed evidence record explicitly authorises a measured statement.

#	Non-claim
1	No measured ride-comfort percentage or ISO 2631-1 improvement until L1/L2 evidence
2	No certification, type approval or regulatory compliance of vehicle or module
3	No crashworthiness or survivability demonstration
4	No patent grant, novelty assertion or freedom-to-operate
5	No production-ready mass, cost or schedule commitment
6	No continuous public-road 250 km/h capability
7	No closed mass or energy budget suitable for production release
8	No clinical outcome claim for ambulance use
9	No continuous RMS force capability until thermal bench data
10	No supplier or OEM endorsement
11	No FIA homologation or series eligibility from FIA-referenced design philosophy
12	No detachable parachute capsule in this baseline

Sales language may describe architecture and modes. Sales language may not invent numbers. Option fees and license prices are bilateral term-sheet matters, not DOC-00 published prices.

15. Change Control Against Prior Revisions

R5.8 inherits R5.7 locked hosts/powertrains and adds Contract Diligence §22 (Option, deliverables, figures, REQ-M21 freeze, IP line, reviewer, sales boundary). R5.7 added geometry/powertrain locks (§21). Prior revisions restored and added: §8 SIM-001 skeleton with bands, ladder and SIM-C criteria; §9 gates G0–G5, Ph0–Ph3 with Month 2 ICD procurement and 4–8 month lead, risk register R-01...R-16 covering all referenced risks; continuous numbering including §1.3; T0–T3 definitions aligned to the document tree; Hold function definition for REQ-M16; REQ-M09a status TARGET until STR-001; module mass line-visible table; vehicle WBS mapped to upper empty ≈2,250 kg and consistent hybrid powertrain lines; REQ-M25 horizontal restraint; clearance/stroke stack with energy sensitivity example; value statement and KPI table; legal owner clarity; ISO 4413; FTO non-claim. No R5.3 technical intent was intentionally deleted; compressed material was expanded where the critique identified gaps.

16. Preliminary HARA Expansion and Safety Process Notes

The screening HARA in §6 is expanded here with process expectations for G2. Each candidate hazard shall eventually own: hazardous event description, operational situation, S/E/C rationale, safety goal, safe state, and fault-tolerant time interval where applicable. ASIL assignment is not made in T1.

Hazard	Safe state (draft)	Notes
Loss of geometric capture	Mechanical capture retained; motion inhibited	FTA required
Uncontrolled vertical force	Force bounded; S3 or S4	Command integrity
Loss of static support	Stops + remaining paths carry load	Hydraulic burst cases
Asymmetric lock under lateral load	SPORT not declared; COMFORT or limp	Per-lock sensing
Unlock under high lateral-g	Unlock inhibited; driver informed	REQ-M21 thresholds
Height drop at speed	Fail-closed valves; controlled degrade	Leak detection
Post-event re-enable	S7 holds until manual clear	No auto S2
Maintenance powered motion	S8 physical lockout	REQ-M15

ISO 26262 and ISO 21448 remain candidate process frameworks. Buyer programmes may impose additional standards; those become contract annexes and do not silently rewrite this baseline.

17. Supplier and ICD Outline (Planning)

17.1 Supplier questionnaire themes

Domain	Questions the supplier must answer
Hydraulic macro	Cylinder force at pressure; seal life; fail-closed valve response time; orifice hold option capability
Lorentz / actuator	Km measured; continuous and peak force; thermal map; gap tolerance; mass
Lock	Engage/release time under load; status sensing; shear capacity; contamination tolerance
Stops	Force-displacement curve; energy rating; temperature sensitivity; inspection interval
Sensors	Stroke, force, pressure, temperature accuracy and fault behaviour
Electronics	48 V drive capability; current limit; diagnostic coverage; EMC class

17.2 ICD-001 minimum chapter list

Mechanical envelope and hard-points; bolt patterns and torque class; hydraulic ports and fluid spec (ISO 4413 path); electrical connector pinout and 48 V limits; communication protocol and timing; lock status bits; fail-safe defaults; service clearances; environmental sealing targets; nameplate and serialisation. ICD draft is a G1 artefact; frozen ICD is required before long-lead procurement commitment.

17.3 Configuration management note

Until a formal CM system is established, revision identity is carried by document ID and date on the cover. Child documents (SIM, STR, ICD) shall reference the parent DOC-00 revision they were generated

against. Conflicting numbers between child and parent are resolved by formal change, not by informal email.

18. Master Position Statement

R5.8 is the Contract Diligence Baseline for ATFSV-MOD-001: inherits R5.7 locked hosts and architecture; adds Option template, deliverables register, controlled schematics, REQ-M21 temporary freeze, IP filing line, reviewer targets, and explicit DOC-00 sales boundary (§22). It defines mechanically captured macro–micro isolation with COMFORT and SPORT modes, owned requirements REQ-M01 through REQ-M25, SIM-001 skeleton, gates G0–G5, phase schedule Ph0–Ph3, risk register R-01 through R-16, module-visible mass table, vehicle WBS mapped to upper-body accounting, horizontal restraint requirements, clearance stack energy sensitivity, value KPIs without invented comfort percentages, and commercial Path A engagement without published fixed licence prices. TRL remains 2–3. Legal owner is Amirbahador Atai. No test result, certification, patentability or FTO is asserted.

The remaining work is evidence and formalisation: execute SIM-001, run single-tower bench, freeze ICD and STR-001, appoint independent reviewer, and pursue counsel-led IP filings. The architecture is sufficiently specified for supplier engagement and controlled NDA diligence.

19. Dual Host Variants V-L and V-H — Closed Mass Budgets, Carbon Cell, FIA-Referenced Roll

R5.8 freezes two host-vehicle mass variants that share the same CMFSB module architecture (four-tower interface, COMFORT/SPORT). Both variants specify a full carbon-fibre passenger cell in the design philosophy of high-end monocoque cockpits (Bugatti/Pagani class stiffness and craftsmanship as qualitative benchmarks only — no copy, no trademark claim, no performance equivalence claim) and a carbon / hybrid roll structure designed to the load philosophy of FIA competition survival cells. **No FIA homologation, passport, or series eligibility is claimed.** Detachable parachute recovery remains outside this baseline.

19.1 Variant definitions

Variant	Role	GVW hard ceiling	Occupants	Module load class
V-L Urban Light	Road / limited urban & grand-touring host	4,000 kg	4 design	S (light corner)
V-H Military / Off-road	Specialty, utility, defence-light, extreme	7,000 kg	Up to 6	M / L

19.2 Carbon passenger cell (both variants)

Both V-L and V-H use a closed structural carbon-fibre cell as the primary upper body. The cell carries seats, restraints, and roll-structure members and mounts to the four CMFSB hard-points. Design philosophy: high torsional stiffness, continuous load paths, integrated belt and seat hard-points — analogous in intent to hypercar carbon cockpits, not a replica of any production car.

Parameter	V-L	V-H	Status
Material philosophy	CFRP structural cell	CFRP structural cell (heavier laminate / inserts)	TARGET
Occupant capacity	4	Up to 6	BASELINE class
Interface to module	4 hard-points	4 hard-points	BASELINE
Cell mass budget (structure only)	≤280 kg	≤420 kg	TARGET budget
Crash / intrusion role	With roll members + stops	With roll members + stops	STR-001

19.3 FIA-referenced carbon roll structure (both variants)

Both variants integrate roll-over and intrusion members consistent with FIA survival-cell / roll-cage load philosophy (edition to be frozen with structures lead and counsel). Implementation may be integral CFRP members and/or hybrid metallic inserts at belt and mount points. This is a design reference — not an FIA homologation claim.

Item	V-L	V-H
Roll structure mass budget	≤45 kg (integral + inserts)	≤70 kg
Load cases	Roof, side, belt anchorage — values OPEN until edition freeze	Same + higher utility load share
Homologation	Not claimed	Not claimed

19.4 Closed mass budget — V-L Urban Light (GVW ≤ 4,000 kg)

All lines STUDY P50 for programme decision. Carbon cell + FIA-referenced roll are included. Isolation is class S (lighter full-corner build than V-H).

WBS line	kg	Notes
CFRP passenger cell (structure)	280	§19.2 budget
FIA-referenced roll members / inserts	45	§19.3
Closures + glass + exterior panels	220	Light closures
Interior + 4 seats + belts + HVAC + cabin LV	200	4-occupant
Lower chassis + crossmembers	320	4×4
E-drive / small hybrid power unit	200	Urban class
Battery 40–55 kWh class	320	Not 130 kWh
HV/LV distribution + cooling	70	
Axles / steering / foundation suspension	240	
Wheels + tyres + brakes	200	
Isolation 4× class-S corners	280	CMFSB light build
Fluids + fillers	50	
Programme reserve	75	
Curb (no occupants)	≈2,500	Sum of lines
4 occupants × 90 kg	360	Design case
Urban cargo allowance	100	
GVW design case	≈2,960	Margin to 4,000
Hard ceiling	4,000	Shall not be exceeded in claims

V-L is feasible only with 4 occupants, modest energy storage, class-S isolation, and strict control of exterior/interior mass. A 6-seat luxury interior or 130 kWh battery breaks the 4 t ceiling.

19.5 Closed mass budget — V-H Military / Off-road (GVW ≤ 7,000 kg)

WBS line	kg	Notes
CFRP passenger cell (structure)	420	Heavier laminate / 6-seat hard-points
FIA-referenced roll members / inserts	70	§19.3
Closures + glass + exterior (reinforced)	380	
Interior + up to 6 seats + belts + HVAC + LV	320	
Lower chassis + crossmembers (reinforced)	650	4×4 default; 6×6 optional trade
Hybrid ICE + generator class	400	
Traction motors + reductions	350	
Battery 70–90 kWh class	500	
HV/LV + cooling heavy	140	
Axles / steering / driveline	420	

WBS line	kg	Notes
Wheels + off-road tyres + brakes	360	
Isolation 4× class-M/L corners	380	CMFSB full corner class
Light protection package (optional)	200	If omitted → reserve
Fluids + partial fuel	180	
Programme reserve	150	
Curb (no occupants)	≈4,920	Sum of lines
6 occupants × 90 kg	540	Design case
Mission payload allowance	400	
GVW design case	≈5,860	Margin to 7,000
Hard ceiling	7,000	Shall not be exceeded in claims

19.6 Shared rules for both variants

Rule	Statement
Module IP	Same CMFSB architecture; Path A license does not require buyer to take both variants
Carbon cell	Mandatory design intent on both V-L and V-H within stated mass budgets
FIA reference	Load philosophy only — zero homologation claim in T1 materials
First pilot	4×4 recommended; V-L preferred for fastest road pilot; V-H when buyer funds utility path
Prior 7.5 t single story	Superseded as the sole commercial mass story; use V-L / V-H ceilings
6×6	Optional on V-H only if axle mass traded from reserve/protection

19.7 Requirements additions

ID	Requirement	Gate
REQ-V01	V-L GVW claims shall remain ≤4,000 kg including stated occupant policy	G1
REQ-V02	V-H GVW claims shall remain ≤7,000 kg including stated occupant/payload policy	G1
REQ-V03	Both variants implement CFRP passenger cell within \$19 mass budgets	G2
REQ-V04	Both variants implement FIA-referenced roll structure; no homologation claim without formal process	Ongoing
REQ-V05	External materials name V-L and/or V-H explicitly when quoting vehicle mass	Ongoing

20. License-Engagement Package — Path A Document Set and Readiness Boundary

This section defines what a Path A module license engagement is expected to use as the controlled document set, how V-L and V-H host documents relate, and the honest readiness boundary of DOC-00 at R5.8. It does not publish licence prices and does not assert that evidence gates G2–G5 are already closed.

20.1 Recommended document volume (answer to 20 + 25 + 25)

Scope	Recommended controlled count	Comment
Path A — module core	≈12–20 master documents	Sufficient for NDA, Option, and first licence term sheet
V-L host vehicle	≈15–25 if buyer takes vehicle path	Not required for module-only Path A
V-H host vehicle	≈15–25 if buyer takes vehicle path	Not required for module-only Path A
Total if module + both hosts fully documented	≈50–70	Programme scale — not a prerequisite to open Path A Option
80+ documents at day one	Not recommended	Slows first contract; build after Option or funded pilot

A Path A licence can close against the module core set. Host-vehicle document packs are sold or funded only when the buyer selects V-L and/or V-H integration.

20.2 Path A module core list (planning — up to 20)

#	Document	Role	Status at R5.8
1	DOC-00 (this baseline)	Architecture, REQ, risk, variants	Issued T1
2	MOD-001 Module datasheet	1–3 page buyer summary	To issue from DOC-00
3	ICD-001 Interfaces	Mech/elec/comm/safety/lock	Outline defined; draft G1
4	SIM-001 Simulation	Comfort vs Sport evidence L1	Skeleton issued; run OPEN
5	STR-001 Structures / stops	Stop curve, horizontal loads	Required G2
6	LCK-001 Lock system	SPORT lock geometry & status	Required G2
7	ACT-001 Actuator / thermal	Force, Km, duty cycle	Required G2
8	HYD-001 Hydraulics	Macro stage, valves, ISO 4413 path	Required G1/G2
9	SAF-001 HARA / FMEA pack	Expanded safety work products	Screening in DOC-00; formal G2
10	TST-001 L2 bench plan	Single-tower acceptance	Outline in App C
11	TST-002 L3 HIL plan	Four-corner / mode tests	Outline in App C
12	CFG-001 Configuration index	Revision & hash register	To establish
13–20	Optional: EMC, materials, supplier specs, CAL, BOM study, quality, service, training	As needed under Option	Not day-one blockers

20.3 Host variant packs (up to ≈25 each — only if Path B / vehicle sale)

Examples (each of V-L / V-H)	Purpose
Geometry & hard-points	Wheelbase, length, tower layout
Mass properties & WBS	Variant budget control vs 4 t / 7 t
Carbon cell design note	Laminate, joins, 4- or 6-seat
Roll structure note	FIA-referenced load cases
Powertrain / energy	Battery class, hybrid, range study
Brake & tyre	Fade and rating for mass class
Electrical architecture	48 V module vs vehicle HV
Homologation map	Market category — OPEN by region
Interior / HMI modes	SPORT/COMFORT driver interface
Off-road / protection (V-H)	Only V-H as applicable
... up to ≈25 controlled titles	Built after module Option or funded vehicle pilot

20.4 What is included in a Path A licence engagement (planning)

Typically included	Not included unless separate deal
Rights to use CMFSB module architecture in a named field	Assignment of all IP worldwide
DOC-00 + agreed subset of Path A core set under configuration control	Full V-L and V-H vehicle certification packs
Agreed engineering support hours	Unlimited R&D; or production tooling
Dual-mode COMFORT/SPORT behaviour as baselined	Measured comfort % without L1/L2 evidence
Access path to Option → milestones (ICD, bench, support)	Guaranteed regulatory approval

20.5 Readiness boundary of R5.8 (honest)

Use case	R5.8 suitable?	Condition
NDA + T1 technical diligence	Yes	This document under NDA
Paid Option negotiation	Yes	Bilateral term sheet; fee not published here
Module licence term sheet	Conditional	Prefer ICD draft + SIM or bench plan committed
Large upfront assignment / exclusive sale	Not on DOC-00 alone	Requires IP filing status, FTO counsel, evidence
Production / type approval claim	No	Explicit non-claim
Advertising measured ride comfort	No	Until L1/L2 configuration-managed results

R5.8 is therefore designated a **License-Engagement Baseline**: complete enough to open and structure a Path A commercial path, not a certificate of finished product. Major architectural and mass-variant decisions are frozen for engagement; residual OPEN items are listed in requirements and risk registers and are normal at TRL 2–3.

20.6 Residual OPEN items that do not block NDA (but block exaggerated claims)

Numeric freeze of REQ-M21 lateral-g policy; STR-001 stop curve values; measured Km; formal HARA ASIL; independent reviewer signature; provisional patent numbers; supplier-weighted mass. These are work items under G1–G2, not reasons to withhold T1 under NDA.

21. Locked Host Geometry, AMG Hybrid Powertrains and Closed Mass Budgets (V-L / V-H)

R5.8 freezes host-vehicle external geometry, hybrid powertrain selection, fuel capacity, battery energy, range target, and revised mass budgets for both variants. These values supersede any earlier TBD dimension notes and any prior pure-electric energy lines for V-L/V-H host accounting. The CMFSB module architecture (four-tower interface, COMFORT/SPORT) is unchanged. All power and consumption figures remain TARGET/STUDY until tested.

21.1 Geometry lock — both variants

Parameter	V-L Urban Light	V-H Military / Off-road	Status
Overall length	5,000 mm	5,500 mm	LOCKED
Overall width (mirrors excluded)	2,000 mm	2,200 mm	LOCKED
Overall height (SPORT / seated)	1,700 mm	2,000 mm	LOCKED
Wheelbase	3,000 mm	3,300 mm (front axle to mid of rear tandem)	LOCKED
Ground clearance (SPORT)	200 mm	300 mm	LOCKED
Wheels / drive	4x4 (four wheels)	6x6 — one front axle + dual rear tandem	LOCKED
Design occupants	4	6	LOCKED
Passenger cell	Full CFRP structural cell	Full CFRP structural cell	BASELINE
Roll structure	FIA-referenced CFRP/hybrid	FIA-referenced CFRP/hybrid	Reference only — no homologation claim
GVW hard ceiling	4,000 kg	7,000 kg	LOCKED

Six-wheel layout for V-H is fixed as single front axle + dual rear axles (tandem). Equal three-axle spacing and dual-front layouts are not baseline.

21.2 Powertrain lock — hybrid AMG-class ICE + electric assist

Both variants are hybrid. Marketing may reference AMG-class eight- and twelve-cylinder families as the ICE performance benchmark; final supplier part numbers are OPEN. No OEM endorsement is claimed.

Parameter	V-L	V-H	Status
Architecture	Hybrid (ICE + e-drive)	Hybrid (ICE + e-drive)	LOCKED
ICE family	8-cylinder AMG-class (4.0 biturbo class)	12-cylinder AMG-class (6.0 biturbo class)	TARGET family
ICE power (planning)	≈430–450 kW class	≈460–500 kW class	TARGET
Electric traction (planning)	≈150–180 kW total	≈200–250 kW total	TARGET
Battery usable energy	20 kWh	40 kWh	LOCKED
Fuel tank	80 L	150 L	LOCKED
Combined range target	1,000 km	1,000 km	TARGET mission — not tested
Implied fuel intensity (planning)	≈8 L/100 km combined class	≈15 L/100 km combined class	STUDY for 1,000 km on tank

21.3 Closed mass budget — V-L (hybrid V8, 20 kWh, 80 L)

WBS line	kg	Notes
CFRP passenger cell	280	4-seat hard-points
FIA-referenced roll members	45	
Closures + glass + panels	230	
Interior 4-seat + HVAC + cabin LV	200	
Lower chassis	340	4×4
AMG-class V8 ICE + ancillaries	275	8-cylinder hybrid ICE
Hybrid transmission / coupling	160	
E-motors + inverters	130	
Battery 20 kWh usable	145	LOCKED energy
HV/LV + cooling	95	
Axles / steering	260	
Wheels + tyres + brakes	210	
Isolation 4× class-S corners	280	CMFSB
Fuel tank 80 L + reference fuel	75	
Exhaust + fluids	70	
Programme reserve	70	
Curb	≈2,865	Line sum
4 × 90 kg occupants	360	
Urban cargo	100	
GVW design case	≈3,325	Margin to 4,000
Hard ceiling	4,000	LOCKED

21.4 Closed mass budget — V-H (hybrid V12, 40 kWh, 150 L, 6×6)

Battery usable energy for V-H is locked at **40 kWh** (updated from earlier 32 kWh study). Mass line uses ≈260 kg for the 40 kWh pack class.

WBS line	kg	Notes
CFRP passenger cell	420	6-seat hard-points
FIA-referenced roll members	70	
Closures + glass + reinforced exterior	400	
Interior 6-seat + HVAC + LV	330	
Lower chassis reinforced	720	6×6 structure
AMG-class V12 ICE + ancillaries	390	12-cylinder hybrid ICE
Hybrid transmission / coupling	190	
E-motors + inverters	180	
Battery 40 kWh usable	260	LOCKED energy
HV/LV + heavy cooling	160	
Three-axle 6×6 driveline	580	Front + dual rear
Wheels + off-road tyres + brakes	480	Six corners
Isolation 4× class-M/L corners	380	CMFSB
Light protection package (optional)	200	Else → reserve
Fuel tank 150 L + reference fuel	140	
Fluids	100	
Programme reserve	150	
Curb	≈5,150	Line sum with 40 kWh
6 × 90 kg occupants	540	

WBS line	kg	Notes
Mission payload	400	
GVW design case	≈6,090	Margin to 7,000
Hard ceiling	7,000	LOCKED

21.5 Cross-check and non-claims for locked hosts

Check	Result
V-L design GVW vs 4,000 ceiling	≈3,325 < 4,000 — margin retained
V-H design GVW vs 7,000 ceiling	≈6,090 < 7,000 — margin retained
1,000 km range	TARGET from tank + hybrid management — not road-test evidence
AMG naming	Class/family benchmark only — no OEM partnership claimed
Module Path A	Independent of host ICE choice; buyer may license module only

21.6 Requirements additions for locked hosts

ID	Requirement	Status
REQ-V06	V-L external geometry shall match §21.1 locked dimensions	LOCKED
REQ-V07	V-H external geometry shall match §21.1 locked dimensions and 6×6 tandem-rear layout	LOCKED
REQ-V08	V-L powertrain shall be hybrid with 8-cylinder AMG-class ICE and 20 kWh usable battery	LOCKED
REQ-V09	V-H powertrain shall be hybrid with 12-cylinder AMG-class ICE and 40 kWh usable battery	LOCKED
REQ-V10	Fuel capacity V-L 80 L / V-H 150 L; combined range target 1,000 km both	TARGET range

22. Contract Diligence Addendum — Option, Deliverables, Schematics, REQ-M21, IP, Reviewer

Section 22 adds the seven engagement items required to structure a serious Path A licence discussion without deleting any prior architecture, table, or requirement. It does not assert that G2 bench evidence or patent grants already exist.

22.1 Path A Option term sheet (planning template — non-binding until signed)

Numeric fees are bilateral and are intentionally not published as fixed public prices in DOC-00. The structure below is the baseline offer shape for counsel to convert into a definitive Option agreement.

Term	Baseline planning value	Notes
Option purpose	Exclusive evaluation of Path A module licence in a named field of use	Does not grant production licence until exercised
Option duration	90 days from effective date (extendable once by mutual written agreement)	TARGET structure
Option fee	Paid fee — amount negotiated (not stated as a public fixed USD figure herein)	COMMERCIAL OPEN
Fee credit	Up to 100% of Option fee may credit against first licence payment if licence closes within Option term + 30-day grace	Negotiable
Exclusivity	Limited exclusivity for named field / territory during Option only	Optional; wider exclusivity costs more
T2 access under Option	ICD drafts, SIM assumptions, lock interface detail, detailed WBS as need-to-know	Under NDA + Option
Buyer diligence rights	Technical questions, one structured review meeting, written clarification responses	Hours capped in Option
No licence obligation	Neither party is obliged to close a licence solely because Option was signed	BASELINE principle
IP ownership during Option	Remains with Legal Owner (Amirbahador Atai) unless separate assignment is signed	BASELINE
Governing law / venue	To be agreed in Option (counsel)	OPEN
Expiry outcome	T2 access ends; T1 may remain under residual NDA terms	

22.2 Path A deliverables register (configuration status)

ID	Deliverable	Status at R5.8	Target gate
D-01	DOC-00 Master baseline (this document)	Issued R5.8	G0
D-02	MOD-001 Module datasheet (1–3 pp)	Due — extract from DOC-00	G1
D-03	ICD-001 Interface control	Outline defined; draft due	G1
D-04	SIM-001 Simulation pack	Skeleton issued; numerical run OPEN	G1
D-05	STR-001 Stops & horizontal loads	Required	G2
D-06	LCK-001 Lock system	Required	G2
D-07	ACT-001 Actuator / thermal	Required	G2
D-08	HYD-001 Hydraulics (ISO 4413 path)	Required	G1/G2
D-09	SAF-001 HARA/FMEA formal pack	Screening in DOC-00	G2
D-10	TST-001 L2 single-tower plan + records	Plan outline in App C	G2
D-11	TST-002 L3 four-corner / HIL plan	Plan outline in App C	G3
D-12	CFG-001 Configuration index	To establish	G1
D-13	Option agreement (commercial)	Template §22.1 — counsel	Commercial
D-14	Licence term sheet (if exercised)	After Option / evidence	G5

22.3 Controlled schematic descriptions (Figure set)

Until CAD figures are configuration-managed, the following textual schematics are the controlled figure set for T1. They define intent for ICD and SIM authors.

Figure 1 — Layer stack (top → bottom)

Layer	Content	Load role
L5	Occupants, seats, restraints, interior	Payload
L4	CFRP passenger cell + FIA-referenced roll members	Structure / crash path with stops
L3	Micro stage (carriage + Lorentz) ± 7.5 mm	Dynamic force only — not sole static path
L2	Macro stage (hydraulic ± 50 mm, leveling, lock seats)	Static support + large travel
L1	Lower chassis / axles / powertrain	Road loads

Figure 2 — Macro–micro force path (one corner)

Path	Elements	Rule
Static / quasi-static	Cell hard-point → capture → macro spring/hydraulic → chassis	Always available
Dynamic isolation	Micro Lorentz + optional damping within stroke	Bounded force; not sole support
SPORT	Seat faces + positive locks → high stiffness path	Confirmed locks only
Crash / overload	Progressive stops + cell + chassis	Independent of closed-loop magnet force

Figure 3 — State machine (summary graph)

From → To	Trigger	Interlock
S1 → S2	Self-test pass	Capture OK; supply OK
S2 ↔ S2S	Mode request COMFORT/SPORT	Physical seat/lock preconditions; no free-float
S2/S2S/S3 → S4	Power loss / critical fault	Hardware-dominant; valves fail-closed
S4 → S1 only	Restore + self-test	Forbidden: S4 → S2 direct
Any → S5 → S6	Impact preparation / event	Stops; no magnet energy dependence
S6 → S7	Event complete	No automatic re-enable

Figure 4 — Power-loss timing path

Time	Action
t0	48 V / control power lost or commanded power-cut test
t0+	Leveling valves fail-closed; Lorentz command removed
t0...settle	Upper body settles on passive macro support within envelope
Optional	If stored energy and window allow: prefer lock engagement toward SPORT-safe posture
End state	S4 — captured, passive support, no software aliveness required for primary safety

22.4 REQ-M21 temporary numeric freeze (TARGET — revise after SIM)

Until vehicle dynamics freeze, the following conservative TARGET numbers apply for COMFORT roll / lateral-g policy. They are planning holds for arbitration design, not measured limits.

Parameter	Temporary TARGET	Use
Max steady roll gradient (COMFORT)	≤ 6.0 deg per g lateral at design payload (V-L)	SIM-C2 / HIL comparison
Inhibit automatic COMFORT entry	When $ a_y > 0.40$ g for > 0.5 s (unless driver overrides)	Arbitration
Inhibit SPORT → COMFORT unlock	When $ a_y > 0.35$ g or yaw-rate threshold TBD in ICD	Safety interlock
Driver override	Always available; logged; does not remove capture invariants	HMI
Revision rule	Numbers may only change by formal change after SIM-001 or mule data	CM

REQ-M21 status remains TARGET (temporary freeze). It is no longer a blank OPEN for arbitration coding.

22.5 Intellectual property filing status (external line)

Item	R5.8 statement
Provisional / patent applications	Not yet filed at issue date of R5.8. Upon filing, replace this cell with: application number(s), office, and date.
Distribution rule	Do not distribute T3 claim charts or enabling claim language in T1 packs until counsel confirms filing status.
Ownership	Legal Owner: Amirbahador Atai — assignment to a company requires a separate controlled deed.
FTO / novelty	Not asserted in this document.

22.6 Independent reviewer and approval targets

Role	Name / status	Target date
Author / Legal Owner	Amirbahador Atai — signed issue authority for T1 text	2026-10-03
Independent technical reviewer	To be appointed (external engineer / academic — named individual required before “reviewed” stamp)	Target appointment: within 60 days of R5.8 issue
Approval authority	Not approved for road use, production, or certification credit	Open until programme authority named

22.7 DOC-00 boundary statement (mandatory for sales and diligence)

DOC-00 is a controlled engineering and commercial-structure baseline. It is suitable to support NDA diligence and Path A Option negotiation. It is not, by itself: a certificate of performance; a type approval; a warranty of mass, range, or comfort; a patent grant; or a binding price list. Large upfront payments for full IP assignment require counsel, filing status, and evidence gates beyond this document. Any external presentation that contradicts this boundary is non-controlled and shall be withdrawn.

Appendix E — Release and Sign-Off

Role	Name / appointment	Organisation	Date	Signature
Author / Legal Owner	Amirbahador Atai	ATFSV Engineering	2026-10-03	_____
Independent Reviewer	To be appointed	External / academic / engineering	Pending	_____
Approval Authority	To be appointed	Programme / customer authority	Pending	_____

ISSUED FOR CONTROLLED T1 REVIEW. Independent review and programme approval remain open. Preliminary — not certified — no safety credit without verification evidence.

END OF DOCUMENT — ATFSV-DOC-00-R5.8 — CONTRACT DILIGENCE BASELINE | Legal Owner: Amirbahador Atai | 2026-10-03